

Lab 3 Report

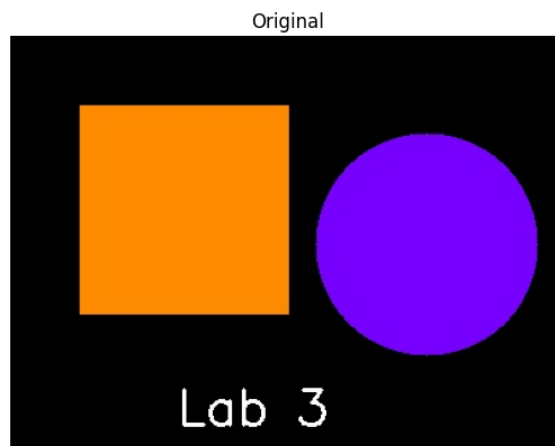
Image Manipulations using OpenCV

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This report covers the two tasks from the lab: geometric transformations (resize, rotate, shear) and intensity transformations (negative, log, power law). All code was run in Python using OpenCV, NumPy, and Matplotlib. The Jupyter notebook and the image used are included in the submitted GitHub folder.

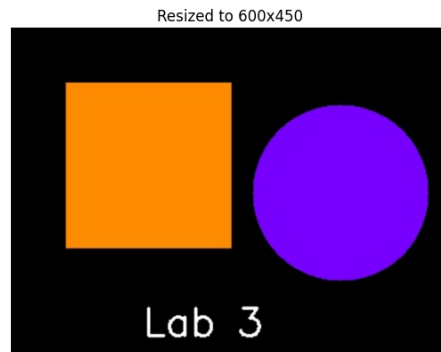
Original image



Task 1: Geometric Transformations

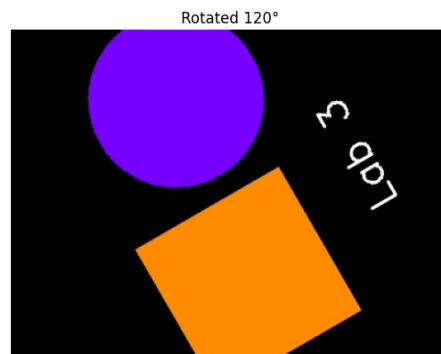
1. Resize (increase image size)

The image was scaled up by 1.5x using `cv2.resize` with linear interpolation.



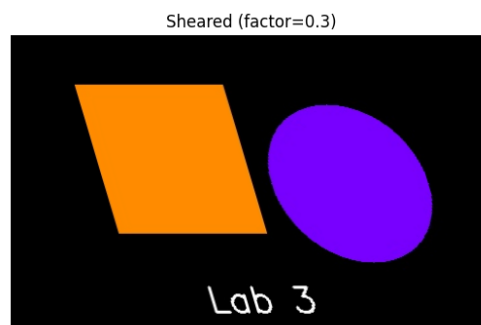
2. Rotate by 120 degrees

`cv2.getRotationMatrix2D` was used to rotate the image 120 degrees around its center.



3. Shear

An affine shear matrix (shear factor 0.3 along x) was applied with `cv2.warpAffine`.



Task 2: Intensity Transformations

1. Negative image

Computed as $s = 255 - r$ for every pixel, inverting the image intensities.

Negative

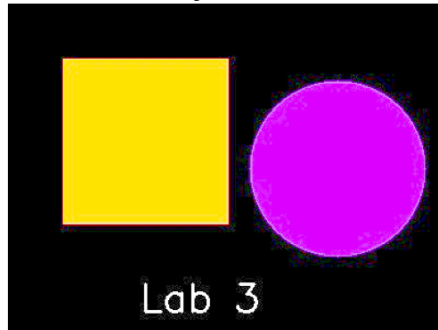


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2. Log transformation

$s = c * \log(1 + r)$, with c scaled so the brightest pixel maps to 255.

Log Transform

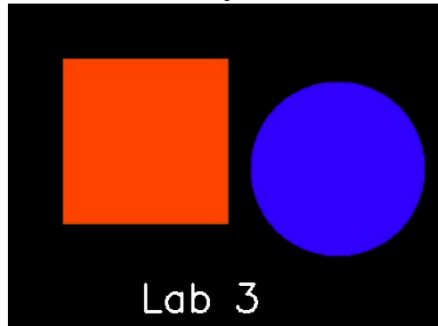


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3. Power law (gamma) transformation

Gamma = 2.2 was applied on the normalized image ($s = r^{\text{gamma}}$) to increase contrast.

Power Law (gamma=2.2)



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